AI-Driven Academic Risk & Dropout Early-Warning System

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Abstract

Traditional academic systems mainly identify weak students after semester or examination results and treat student information as static records. This limits continuous monitoring and early intervention for academically struggling students.

Our proposed Digital Class Twin continuously collects student attendance, assessment, quiz and behavioral data as events. It maintains an evolving digital representation of each student's academic state over time.

Machine learning techniques analyse these data to identify and classify students based on their academic risk.

Problem Statement

Existing Problems

- Weak students are often identified only after examination/semester results.
- Student data is commonly maintained as static records.
- Static records do not adequately represent changing student performance over time.
- Rule-based approaches such as attendance thresholds cannot capture the complete academic situation.
- Institutions lack a mechanism for continuous monitoring and early intervention.

Need for the project

To design a system that can continuously monitor student progress, identify declining academic students at an early stage, and provide understandable evidence to teachers and counselors.

Objectives

1. To collect student academic data such as attendance, internal marks, assignment marks, lab scores, and behaviour records using APIs and Excel/CSV uploads.
2. To create a digital profile for every student that automatically updates based on the latest 30 days of academic activities.
3. To calculate student performance scores (0-100) using Machine Learning and classify students into three levels: Stable, Warning & Critical.
4. To provide a simulation feature that allows students and faculty to check how improvements in attendance and marks can affect performance without changing actual records.
5. To provide secure, role-based dashboards and automatic alerts for students, faculty mentors, and parents.

Related Sustainable Development Goals (SDG)

PRIMARY SDG

SDG 4: Quality Education

SDG 4

Closest match to our project objective

Early identification of academically at-risk students for timely support

- Risk score from academic performance, attendance and behavior
- Low / Moderate / High levels show who needs help first
- Dashboards let teachers and parents act early
- Aims to improve educational outcomes and reduce dropout

SECONDARY SDG

SDG 10: Reduced Inequalities

SDG 10

- Early identification can help students who need extra academic support
- Supports more equitable access to timely guidance
- A supporting link, not a claim to remove inequality

SECONDARY SDG

SDG 9: Industry, Innovation & Infrastructure

SDG 9

- Applies digital-twin concepts to student academic profiles
- Data-driven analytics and digital dashboards in education
- Rule-based risk engine today; trained ML planned

The project primarily contributes to SDG 4 by supporting early identification of academic risk and timely educational intervention.

Existing System

1. Separate & Fragmented Storage

- **Data Silos:** Attendance, CIA marks, lab scores, and conduct records are kept in separate sheets or ERP modules.
- **No Unified View:** Faculty cannot see a student's complete semester standing in one centralized profile.

2. Delayed Performance Processing

- **Late Detection:** Attendance % and overall grades are calculated only monthly or at the end of the semester.
- **Missed Recovery Window:** Too late to arrange remedial classes or meet the 75% Anna University attendance limit.

3. Static Risk Scores

- **No Explainability (XAI):** Existing tools output raw risk percentages without explaining the root cause.
- **Hidden Volatility:** Fails to distinguish steady learners from students with erratic test score swings.

4. No Interactive What-If Simulation

- **Static Dashboards:** Traditional ERPs only display past records without forward-looking recovery projections.
- **No Actionable Planning:** Students and mentors cannot simulate how many classes or marks are needed to recover.

SI NO.	Author(s) & Year	Paper Title	Dataset / Application	Method / Technique Used	Limitations	Key Findings
1	Cagliero et al. (2021)	Predicting Student Academic Performance by Means of Associative Classification	Learning analytics/course data	• Associative classification • Early exam-success prediction	• Course/exam focused • No continuous Digital Twin or intervention history	• Early data can predict exam success • Supports targeted help for at-risk students
2	Karalar et al. (2021)	Predicting Students at Risk of Academic Failure Using Ensemble Model During Pandemic in a Distance Learning System	Moodle + Adobe Connect	• Ensemble ML • GB, QDA, DT, RF, ET, LR, ANN	• Pandemic/distance-learning context Limited • continuous multi-source monitoring	• Ensemble model identified at-risk students • Asynchronous activity was highly informative
3	Park & Yoo (2021)	Early Dropout Prediction in Online Learning of University Using Machine Learning	University online-learning data	• ML classification • Early dropout prediction	• Online-learning focused • No persistent student-state model or intervention loop	• ML can identify potential dropouts early
4	Dabhade et al. (2021)	Educational Data Mining for Predicting Students' Academic Performance Using Machine Learning Algorithms	Academic, personal and behavioral data	•Educational Data Mining •Comparative ML	•Conventional prediction approach • No Digital Twin or closed-loop intervention	• Multiple student factors support performance prediction

5	Tamada et al. (2022)	Predicting Students at Risk of Dropout in Technical Course Using LMS Logs	• Random Forest, DT, NB • K-means + feature engineering	• 1 million Moodle logs	• RF performed strongly • LMS engagement was useful for early prediction	• LMS/technical-course focused • Broader academic and behavioural events not integrated
6	Yağcı (2022)	Educational Data Mining: Prediction of Students' Academic Performance Using Machine Learning Algorithms	• Educational Data Mining • Supervised ML	• Student academic dataset	• ML identified patterns linked to academic achievement	• Dataset-level prediction • No continuously updated student state
7	Oqaidi et al. (2022)	Towards a Students' Dropout Prediction Model in Higher Education Institutions Using Machine Learning Algorithms	• ML classification • Feature selection + evaluation	• Higher-education student data	Feature selection and model choice affect dropout prediction	• Prediction focused • No Digital Twin or intervention feedback
8	Segura et al. (2022)	Machine Learning Prediction of University Student Dropout: Does Preference Play a Key Role?	• SVM, DT, ANN, LR • Feature analysis	• University of Madrid • 28 variables	• First-semester performance improves dropout detection • Course preference is relevant	• Very-early prediction remains limited • Single institutional context

9	Katsuragi & Tanaka (2022)	Dropout Prediction by Interpretable Machine Learning Model Towards Preventing Student Dropout	• Interpretable ML • Learning+ communication features	• Online-school data • Daily learning + Slack activity	• Behaviour and communication can support dropout prediction	• Online context • No multi-source Digital Twin
10	Su et al. (2022)	Applying Machine Learning Tecnologies to explore Students Learning Features and performance Prediction	• RF, SVM, LR, NN • Correlation analysis	• Programming-course quizzes + system logs	• Learning behaviour correlated with performance • NN performed well	• Programming-course focused • Limited broader academic/behavioural data
11	Plak et al. (2022)	Early Warning Systems for More Effective Student Counselling in Higher Education	• LR, additive LR, SVM, RF • Early-warning + counseling	• VU Amsterdam • Two first-year cohorts	•Prediction can support student-specific counselling	• Institution-specific •Intervention depends on local counselling practice
12	Qin et al. (2023)	Early Warning of Student Performance With Integration of Subjective and Objective Elements	• Neural network • Multi-source data synthesis + cost-sensitive learning	• University records • Academic,library, registration, access-control data	• Combining academic + behavioural factors improved warning	• Institution-specific integration • Deep-model interpretability remains challenging

13	Cho et al. (2023)	A Study on Dropout Prediction for University Students Using Machine Learning	• Machine learning • Dropout prediction + feature analysis	•University student records	• ML identified useful dropout-related factors	• Dropout focused • No continuous state tracking or intervention loop
14	Balcioğlu & Artar (2023)	Predicting Academic Performance of Students with Machine Learning	• DT, SVM, NN • Ensemble model	• OULAD	• Ensemble achieved strong prediction performance	• Performance prediction focused • No persistent twin + XAI intervention workflow
15	Nagy & Molontay (2023)	Interpretable Dropout Prediction: Towards XAI-Based Personalized Intervention	• ML dropout prediction • Explainable AI	•Higher-education student data	• Interpretability can support personalized intervention	• Mainly prediction + explanation • No event-driven Digital Twin trajectory
16	Villar & Robledo Velini de Andrade (2024)	Supervised Machine Learning Algorithms for Predicting Student Dropout and Academic Success: A Comparative Study	• DT, SVM, RF, GB, XGBoost, CatBoost, LightGBM • SMOTE/ADASYN + SHAP	•Higher-education dataset	• Boosting models performed strongly • SHAP supported factor interpretation	• Single-dataset dependence • Minority-class prediction remains challenging

17	Vaarma & Li (2024)	Predicting Student Dropouts with Machine Learning: An Empirical Study in Finnish Higher Education	• ML prediction • Monthly longitudinal feature analysis	• Finish higher education •Transcript + demographic + Moodle data	• Credits, failed courses and Moodle activity were important • Feature importance changes over time	• Single-university confidential data • Requires continual updating
18	Tiukhova et al. (2024)	Explainable Learning Analytics: Assessing the Stability of Student Success Prediction Models by Means of Explainable AI	• Predictive learning analytics • SHAP + model-stability/drift analysis	• Student success prediction data	• SHAP helps assess feature importance and stability •Importance changes across contexts	• Explainability focused • Not a complete early-warning/intervention platform
19	Miranda et al. (2024)	Machine Learning's Model-Agnostic Interpretability on the Prediction of Students' Academic Performance in Video-Conference-Assisted Online Learning	• RF, SVM,GNB • SMOTE + SHAP	• 361 online-learning records	• RF performed strongly • SHAP identified important performance factors	• Small, context-specific dataset • Online/pandemic context limits generalization
20	Alalawi et al. (2025)	An Extended Learning Analytics Framework Integrating Machine Learning and Pedagogical Approaches for Student Performance Prediction and Intervention	•Predictive ML + pedagogical intervention • Web-based learning analytics framework	• Course assessment data • Academic pilot	• Connects prediction with intervention • Web framework supports academics	• Intervention depends on academics • Continuous multi-source event integration remains open

Literature Survey , Summary and Research Gap

Research Gap

1. Many studies use static or institution-specific datasets.
2. Most focus mainly on prediction rather than continuous monitoring.
3. Limited integration of academic, attendance and behavioural events.
4. Few systems connect predictions directly with interventions.

Major Methods

1. Machine Learning: Random Forest, Decision Tree, SVM, Logistic Regression
2. Ensemble & Boosting methods for dropout prediction
3. Educational Data Mining and Learning Analytics
4. SHAP / Explainable AI for prediction interpretation
5. Digital Twin & graph-based student pathway modelling

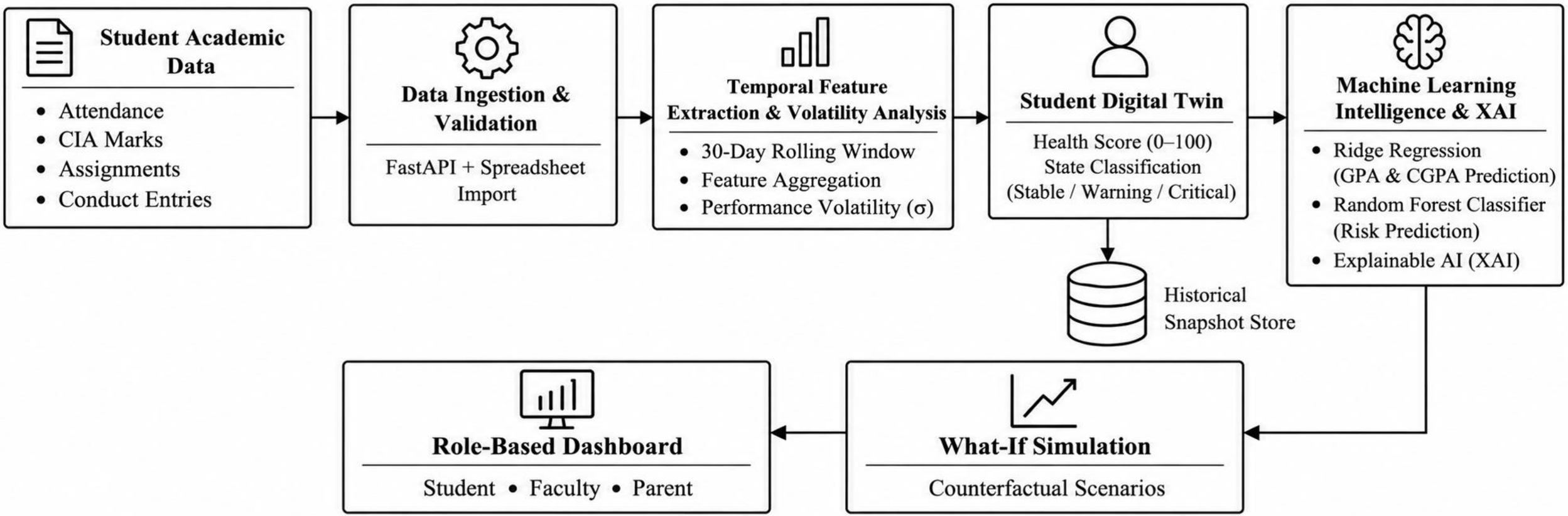
Key Findings

1. ML can predict academic performance and dropout risk at an early stage.
2. Attendance, academic performance, LMS activity and behaviour are important indicators.
3. Combining multiple student data sources improves risk prediction.
4. XAI helps identify the factors influencing predictions.

Relevance to Our Project

1. Our Digital Class Twin extends existing ML and Digital Twin approaches.
2. Uses event-driven continuous monitoring and time-series student state.
3. Provides Low/Medium/High risk, GPA prediction and trajectory-based urgency.
4. Uses SHAP, What-If simulation and intervention logging for actionable decisions.

System Architecture



Module Description

Layer 1: Data Ingestion & File Import

- Collects daily attendance, CIA marks, lab scores, and conduct logs via FastAPI REST APIs.
- Supports bulk Excel / CSV file uploads with automatic column detection.
- Validates data using Pydantic v2 schemas and stores records in SQLite.

Layer 2: 30-Day Rolling Analytics

- Continuously tracks student performance over the last 30 days instead of waiting for end-semester exams.
- Calculates 4 live metrics: Attendance %, Academic Average, Conduct Score, and Grade Volatility.
- Checks a minimum of 3 sessions to avoid false early-semester alerts.

Layer 3: Health Score & State Machine

- Calculates Student Health Score (0-100): 50% Academics + 30% Attendance + 20% Conduct.
- Classifies students into Stable (≥75), Warning (50-74), or Critical (<50).
- Uses a 3.0-point hysteresis buffer to stop repeated false alerts and saves state history.

Layer 4: ML Risk & Grade Prediction

- Random Forest Classifier predicts student dropout risk (*Low, Medium, High*).
- Ridge Regression predicts expected semester SGPA and overall CGPA.
- Explainable AI (SHAP) clearly shows *why* a student is at risk.

Layer 5: What-If Simulation

- Creates a temporary in-memory "Shadow Twin" without changing real database records.
- Lets students and faculty test future attendance % and target exam marks.
- Calculates how many classes or marks are needed to reach the 75% Anna University limit.

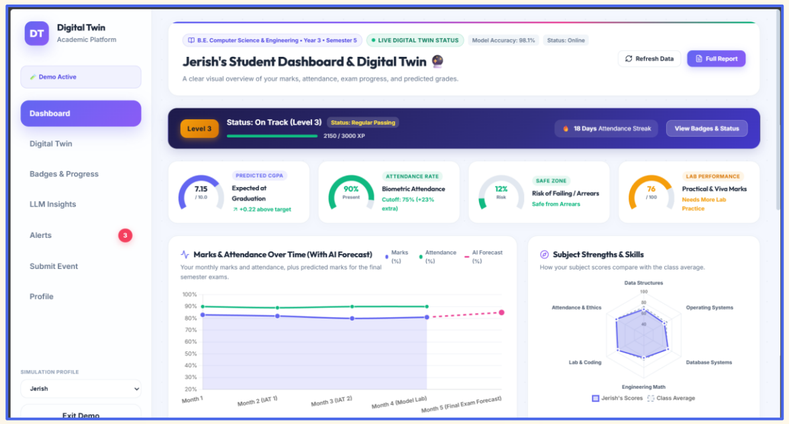
Layer 6: Role-Based Portals & Alerts (FR6-8)

- Interactive React 18 dashboards for Students, Faculty Mentors, and Parents.
- Automatically sends alerts to mentors and parents when a student enters *Warning* or *Critical*.
- Secured with JWT & Bcrypt so parents can view only their own child's record.

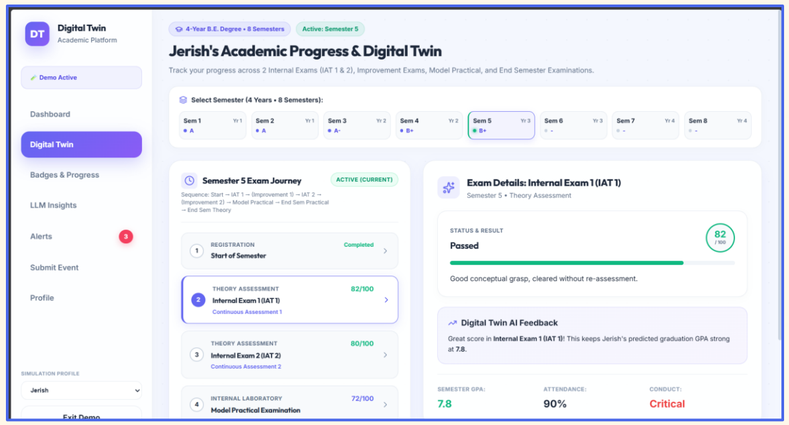
Implementation Details (Framework, Design)

Layer	Technologies
Frontend	React 18
Backend	Python, FastAPI
Database	SQLite 3, SQLAlchemy 2.0
Security	JWT, Role-Based Access Control (RBAC)
ML engine	Random Forest Classifier, Ridge Regression, Explainable AI

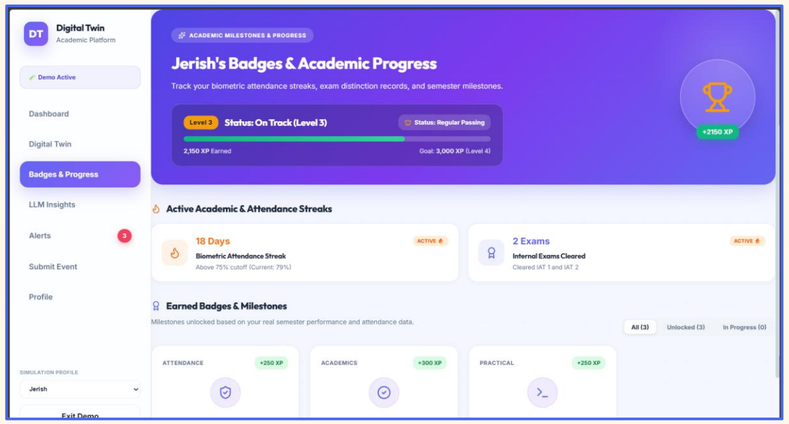
Output Screenshots(Desktop)



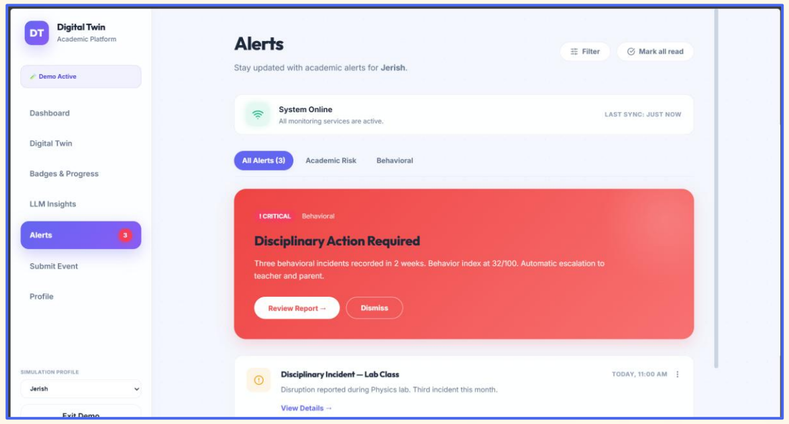
1. Student Dashboard



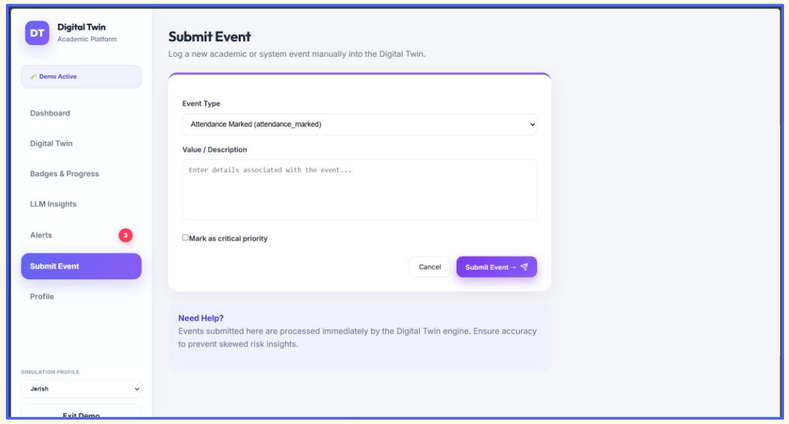
2. Digital Twin - semester progress



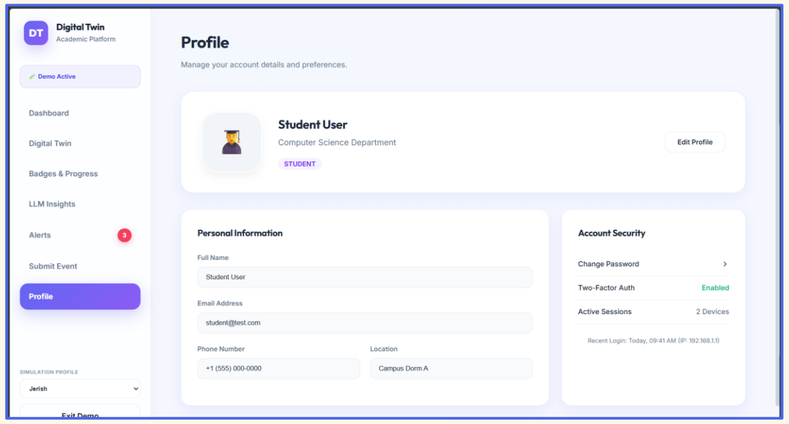
3. Badges & Progress



4. Alerts



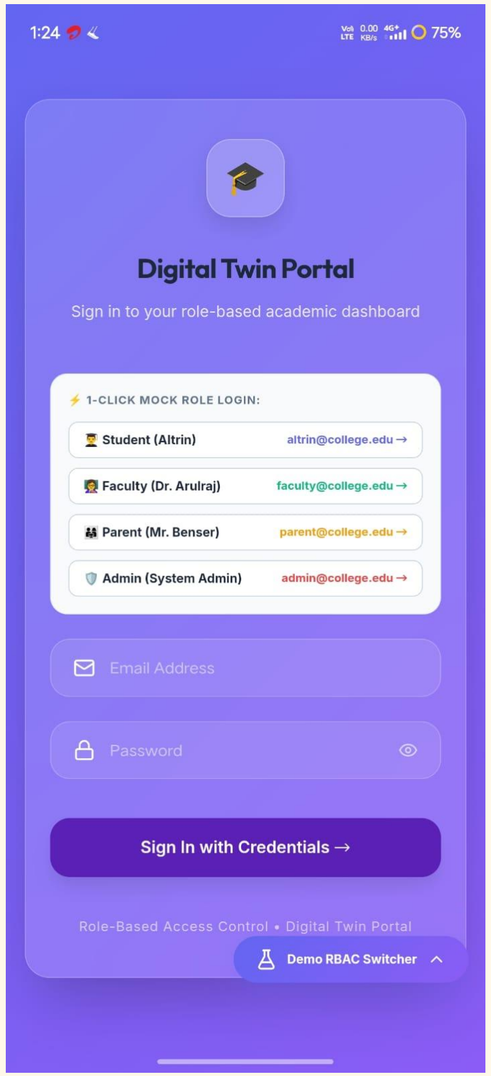
5. Submit Event

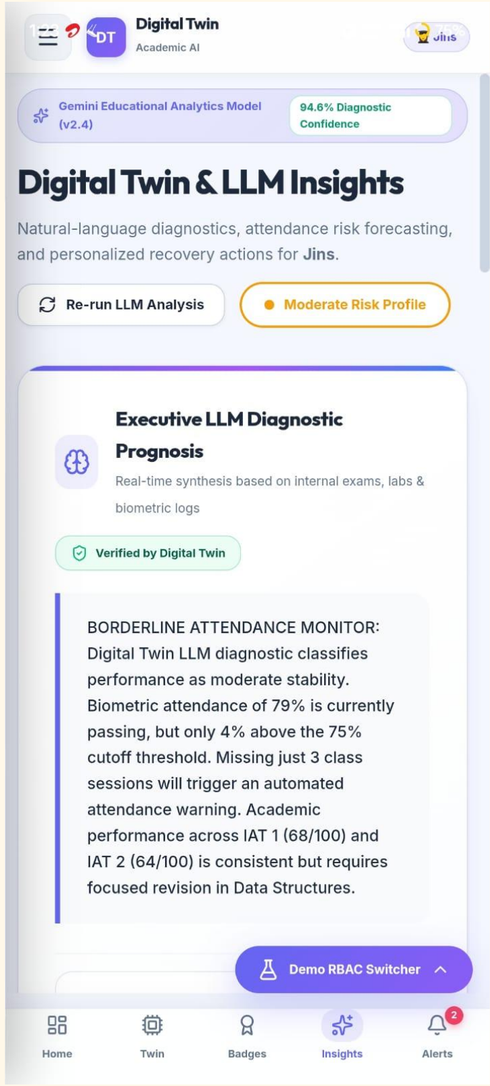


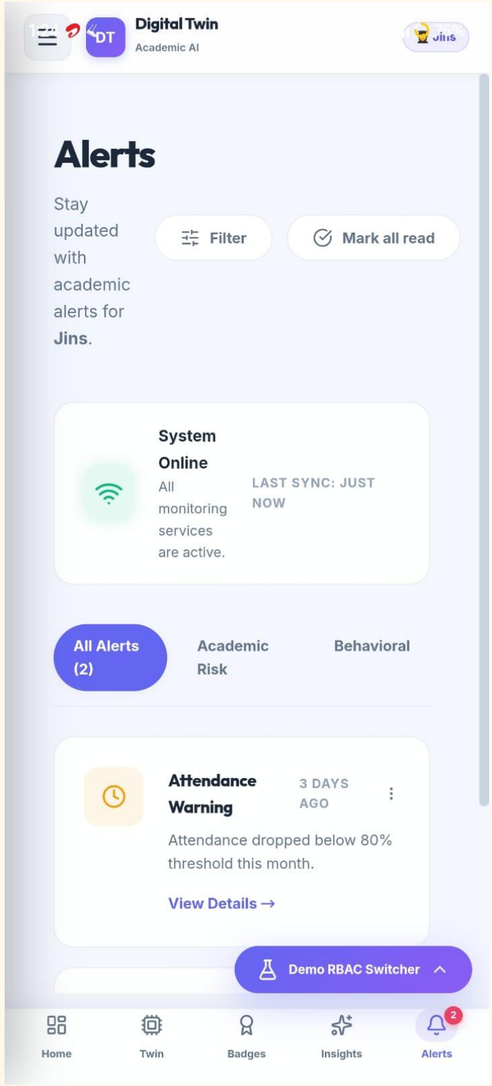
6. Profile

Screenshots from the running prototype in demo / simulation mode.

Output Screenshots(Mobile)







Conclusion

Conclusion

- Built **Digital Class Twin** using **FastAPI, SQLite, and React 18** to identify at-risk students early before end-semester exams.
- Continuously tracks daily attendance, CIA marks, lab scores, and conduct using a **30-day rolling window**.
- Calculates an explainable **Health Score (0-100)** (50% Academic + 30% Attendance + 20% Conduct) and classifies students into **Stable, Warning, or Critical** with a **3.0-point hysteresis buffer**.
- Uses **Random Forest** (risk prediction), **Ridge Regression** (SGPA/CGPA forecast), and a live **What-If Simulator** to help students meet the **75% Anna University attendance limit**.

Future Work

- **Multi-Semester ML Training:** Train **XGBoost & Random Forest** models on multi-year college datasets with **SHAP** explainers.
- **Database & Scalability Upgrade:** Migrate from SQLite to **PostgreSQL** with **Celery & Redis** background task queues for campus-wide deployment.
- **Automated Parent Alerts:** Integrate real-time **SMS, Email, and WhatsApp** alerts for parents and faculty mentors.
- **Mobile App & ERP Integration:** Launch a dedicated **Mobile App** and sync directly with college ERP / LMS platforms.

Participation in Technical Event

Presented at

“International Comference on Advancements in Education, Science, Engineering and Management(ICAESEM 2026)”, B.S.A(P.G) College, Mathura, Uttar Pradesh

October 30-31, 2026

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